PLSC 48110/28110

Lecture 2: Introduction to estimation

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Considering an application with randomization inference.

Loading packages for this class

```
> # install.packages('remotes')
> # library(remotes)
> # install_github('cran/ri')
> library(ri)
> library(ggplot2)
> set.seed(60637)
```

Joint relationships

- Correlation: X and Y are correlated. E.g., $\uparrow$ age and $\uparrow$ income.
- Causation: X *moves* Y. E.g., when I fertilize the plants, I get a higher yield.
  - Implies a *counterfactual*. . If I *hadn't* moved X, we would see a different outcome for Y.
  - Recall Holland (1986): "No causation without manipulation."

Joint relationships: notation

- Dependent variable: (Y), the thing we want to study variation in; also "outcome", "response".
- Independent variable: (W) (or D) in causal settings, the thing that we manipulate. Also "treatment." In non-causal settings, something that may have some natural variation in the population associated with the dependent variable (often X).

Potential outcomes: notation

Assume W is binary.

$$Y_i = \begin{cases} Y_i(0) & W_i = 0 \\ Y_i(1) & W_i = 1 \end{cases}$$

Why is $\mathrm{E}[Y_i(0)]$ different from $\mathrm{E}[Y_i|W = 0]$?
(What is the expectation *over*?)

Potential outcomes: features

1. Associates causal effects with potential outcomes
2. Need to observe outcomes for multiple units (FPoCI!)
3. Role of assignment mechanism

(4.) SUTVA assumption?

Example adapted from:

Chattopadhyay, Raghabendra & Duflo, Esther (2004). Women as policy makers: Evidence from a randomized policy experiment in India.
as discussed in
Gerber, Alan S. & Green, Donald P. (2012). Field experiments: Design, analysis, and interpretation.

Chattopadhyay & Duflo example

- Population: 7 villages
- Treatment: $W = 1$ if female-headed council, $W = 0$ if male
- Outcome: Budget allocation to sanitation

	$Y_i(0)$	$Y_i(1)$	W_i	Y_i	τ_i
Village 1	10	15	1	15	5
Village 2	15	15	0	15	0
Village 3	20	30	0	20	10
Village 4	20	15	0	20	-5
Village 5	10	20	0	10	10
Village 6	15	15	0	15	0
Village 7	15	30	1	30	15
Average	**15**	**20**			**5**

$$\text{ATE} = \mathrm{E}\left[\tau_i\right] = 5$$

Individual treatment effect

$$\tau_i = Y_i(1) - Y_i(0)$$

We're interested in the treatment effect, the difference in potential outcomes had we applied treatment W, vs. if we had not.

- What we can't see: τ_i
- Fundamental problem of causal inference: we can't see *both* potential outcomes for a given unit *at the same time*.
- Holland (1986)'s assumptions: some structure we can impose on the data to back out counterfactuals. (We usually don't know if assumptions are true.)
- Experiments use the independence assumption. . . this one we have some more control over.

Chattopadhyay & Duflo example

What we actually see:

	$Y_i(0)$	$Y_i(1)$	W_i	Y_i	τ_i
Village 1	?	15	1	15	?
Village 2	15	?	0	15	?
Village 3	20	?	0	20	?
Village 4	20	?	0	20	?
Village 5	10	?	0	10	?
Village 6	15	?	0	15	?
Village 7	?	30	1	30	?

We can't directly get τ_i for any individual, and so we also can't take the adverage for τ_i across individuals.

Estimand

The *estimand* is the parameter of interest–it is the quantity that we would like to know about.
For example, when we care about causal effects, the estimand may be:

- the Average Treatment Effect (ATE)
- the Average effect of Treatment on the Treated (ATT)
- the Average effect of Treatment on the Control (ATC)

Why might these three quantities differ?

Estimator

An *estimator* is a function of the data we observe; it is a statistic that gives us an informed guess about the value of the estimand.

Below, the estimator is the function $g(\cdot)$.

$$g(X_1, \ldots, X_n)$$

We can also think of it as a recipe. Given some data, $X_1, \ldots, X_n$, follow the instructions $g(\cdot)$ to produce an **estimate**.

Estimate

An *estimate* is what we calculate from our estimator with a specific set of data. Below, the estimate is the quantity $\hat{\theta}_n$.

$$\hat{\theta}_n = g(X_1, \ldots, X_n)$$



estimand



estimate

Ingredients

- 150g unsalted butter, plus extra for greasing
- 150g plain chocolate, broken into pieces
- 150g plain flour
- ½ tsp baking powder
- ½ tsp bicarbonate of soda
- 200g light muscovado sugar
- 2 large eggs

Method

1. Heat the oven to 160C/140C fan/gas 3. Grease and base line a 1 litre heatproof glass pudding basin and a 450g loaf tin with baking parchment.

2. Put the butter and chocolate into a saucepan and melt over a low heat, stirring. When the chocolate has all melted remove from the heat.

estimator

(image cred: simongrund89)

Independence

- Units that receive treatment look, on average, like units that do not.
- In experiments, we assign treatment *randomly*, so we know that there is no inherent difference between subjects that *could have* received treatment and *could have* received control–all subjects *could have* received all versions of treatment, hypothetically. (The subjunctive mood of research design . . .)
- Why is this so special?
- What this give us: $\mathrm{E}\left[Y_i | W_i = 1\right] = \mathrm{E}\left[Y_i(1)\right]$
- We can back out an estimate of the average treatment effect:

$$\tau = \mathrm{E}\left[Y_i(1) - Y_i(0)\right]$$

Difference-in-means

Proposed estimator for $\mathrm{E}[\tau_i]$: compare people who received treatment 1 to people who received treatment 0.

This is the difference in means estimator:

$$\hat{\tau}_{DM} = \frac{\sum_i^n Y_i W_i}{\sum_i^n W_i} - \frac{\sum_i^n Y_i (1 - W_i)}{\sum_i^n (1 - W_i)}$$

Chattopadhyay & Duflo example

What we actually see:

	$Y_i(0)$	$Y_i(1)$	W_i	Y_i	τ_i
Village 1	?	15	1	15	?
Village 2	15	?	0	15	?
Village 3	20	?	0	20	?
Village 4	20	?	0	20	?
Village 5	10	?	0	10	?
Village 6	15	?	0	15	?
Village 7	?	30	1	30	?

To produce an *estimate* of the ATE, we can compare people who received treatment 1 to people who received treatment 0.

$$\bar{Y}(1) = 22.5 \qquad \bar{Y}(0) = 16$$

$$\hat{\tau} = 22.5 - 16 = \mathbf{6.5}$$

Chattopadhyay & Duflo example

- We have produced an estimate here, by taking the difference in means.
- But what can we say about our uncertainty about this estimate? The number of observations in each group is pretty small.
- Is the estimate meaningfully different from zero?
- How likely would we be to see an effect this size just by chance?

Inference and uncertainty

- Inference is about uncertainty. What is the source of uncertainty here?
- Aside on finite populations vs. random sample from superpopulation.

Chattopadhyay & Duflo example

- One way to think about the magnitude of the effect we observe is to assume the individual treatment effect for all individuals is exactly zero. Then, no matter how we randomized treatment assignment, we would see the same Y_is.
- We can then say how often we would see a treatment effect estimate of this size just by chance, under the assumption that individual treatment effects were actually zero.

Sharp null hypothesis of no effect

We might write the sharp null hypothesis this way:

$$H_0 : \tau_i = 0, \text{ for all } i \text{ in our pop.}$$

This implies that potential outcomes are identical under treatment and control, for all individuals.

$$Y_i(0) = Y_i(1)$$

Sharp null hypothesis of no effect

- To test this hypothesis, we also need to know (or assume we know) the randomization procedure
- Here, we'll assume exactly two villages are assigned treatment.

Chattopadhyay & Duflo example

Then we can fill in this table...

	$Y_i(0)$	$Y_i(1)$	W_i	Y_i	τ_i
Village 1	?	15	1	15	?
Village 2	15	?	0	15	?
Village 3	20	?	0	20	?
Village 4	20	?	0	20	?
Village 5	10	?	0	10	?
Village 6	15	?	0	15	?
Village 7	?	30	1	30	?

Chattopadhyay & Duflo example

. . . as this:

	$Y_i(0)$	$Y_i(1)$	W_i	Y_i	τ_i
Village 1	15	15	1	15	0
Village 2	15	15	0	15	0
Village 3	20	20	0	20	0
Village 4	20	20	0	20	0
Village 5	10	10	0	10	0
Village 6	15	15	0	15	0
Village 7	30	30	1	30	0

Gerber Green example

We can re-run the randomization, and the potential outcomes and observed Y_i will not change, but the treatment effect estimate will.

	$Y_i(0)$	$Y_i(1)$	W_i	Y_i	τ_i
Village 1	15	15	0	15	0
Village 2	15	15	1	15	0
Village 3	20	20	0	20	0
Village 4	20	20	0	20	0
Village 5	10	10	0	10	0
Village 6	15	15	1	15	0
Village 7	30	30	0	30	0

$$\bar{Y}(1) = 15 \qquad \bar{Y}(0) = 19$$

$$\hat{\tau} = 15 - 19 = \mathbf{-4}$$

Gerber Green example

We can re-run the randomization, and the potential outcomes and observed Y_i will not change, but the treatment effect estimate will.

	$Y_i(0)$	$Y_i(1)$	W_i	Y_i	τ_i
Village 1	15	15	0	15	0
Village 2	15	15	0	15	0
Village 3	20	20	1	20	0
Village 4	20	20	1	20	0
Village 5	10	10	0	10	0
Village 6	15	15	0	15	0
Village 7	30	30	0	30	0

$$\bar{Y}(1) = 20 \qquad \bar{Y}(0) = 17$$

$$\hat{\tau} = 20 - 17 = \mathbf{3}$$

Gerber Green example

We can re-run the randomization, and the potential outcomes and observed Y_i will not change, but the treatment effect estimate will.

	$Y_i(0)$	$Y_i(1)$	W_i	Y_i	τ_i
Village 1	15	15	0	15	0
Village 2	15	15	0	15	0
Village 3	20	20	1	20	0
Village 4	20	20	0	20	0
Village 5	10	10	0	10	0
Village 6	15	15	0	15	0
Village 7	30	30	1	30	0

$$\bar{Y}(1) = 25 \qquad \bar{Y}(0) = 15$$

$$\hat{\tau} = 25 - 15 = \mathbf{10}$$

- Because we know how treatment was assigned, we know all the possible ways treatment could be assigned across the villages, and the exact probability.
- There are seven villages, and we say that exactly two will get treatment. Each village is assigned treatment with equal probability.

We can use the package `ri` to find the *exact* distribution of $\hat{\tau}_{DM}$ under the sharp null.

Our real data:

```
> df ← data.frame(
+ # our initial treatment vector
+ W = c(1, 0, 0, 0, 0, 0, 1),
+ # our initial response vector
+ Y = c(15, 15, 20, 20, 10, 15, 30),
+ # treatment assignment probability
+ probs = rep(2/7, 7)
+ )
> df
```

```
  W  Y     probs
1 1 15 0.2857143
2 0 15 0.2857143
3 0 20 0.2857143
4 0 20 0.2857143
5 0 10 0.2857143
6 0 15 0.2857143
7 1 30 0.2857143
```

And our difference in means estimate of the average treatment effect under the real data:

```
> Y1 ← df$Y[which(df$W == 1)]
> Y0 ← df$Y[which(df$W == 0)]
> (dm_hat ← mean(Y1) - mean(Y0))
```

```
[1] 6.5
```

Adding in the hypothetical data.

```
> df ← cbind( # binds the columns together
+ df,
+ # Y(0) under the sharp null of no effect
+ Y0 = df$Y,
+ # Y(1) under the sharp null of no effect
+ Y1 = df$Y)
> df
```

```
  W  Y     probs Y0 Y1
1 1 15 0.2857143 15 15
2 0 15 0.2857143 15 15
3 0 20 0.2857143 20 20
4 0 20 0.2857143 20 20
5 0 10 0.2857143 10 10
6 0 15 0.2857143 15 15
7 1 30 0.2857143 30 30
```

All possible treatment assignments

Consider all the ways treatment could be assigned. This is what `ri` is doing for us.

```
> (perms ← genperms(df$W))
```

```
     [,1]  [,2]  [,3]  [,4]  [,5]  [,6]  [,7]
1       1     1     1     1     1     1     0
2       1     0     0     0     0     0     1
3       0     1     0     0     0     0     1
4       0     0     1     0     0     0     0
5       0     0     0     1     0     0     0
6       0     0     0     0     1     0     0
7       0     0     0     0     0     1     0
```

All possible treatment assignments (continued)

	[,8]	[,9]	[,10]	[,11]	[,12]	[,13]	[,14]
1	0	0	0	0	0	0	0
2	1	1	1	1	0	0	0
3	0	0	0	0	1	1	1
4	1	0	0	0	1	0	0
5	0	1	0	0	0	1	0
6	0	0	1	0	0	0	1
7	0	0	0	1	0	0	0

All possible treatment assignments (continued)

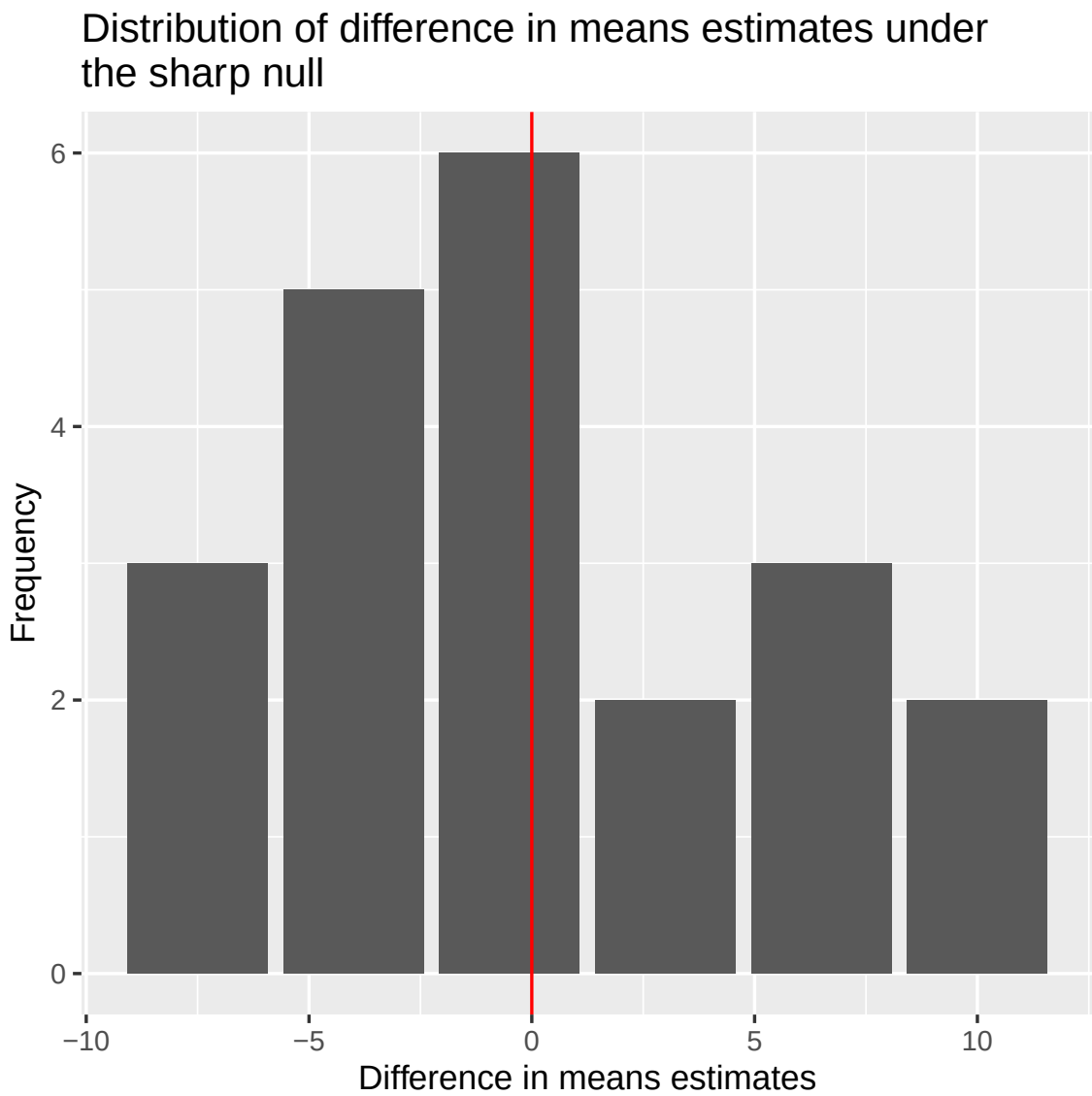
```
  [,15] [,16] [,17] [,18] [,19] [,20] [,21]
1     0     0     0     0     0     0     0
2     0     0     0     0     0     0     0
3     1     0     0     0     0     0     0
4     0     1     1     1     0     0     0
5     0     1     0     0     1     1     0
6     0     0     1     0     1     0     1
7     1     0     0     1     0     1     1
```

Then generate the sampling distribution of the ATE estimate under the sharp null of no effect.

```
> Ys_null ← list(
+ Y0 = df$Y0,
+ Y1 = df$Y1
+ )
> dm ← gendist(Ys_null,
+ perms,
+ prob=df$probs)
> dm
```

```
 [1] -4.0 -0.5 -0.5 -7.5 -4.0  6.5 -0.5
 [8] -0.5 -7.5 -4.0  6.5  3.0 -4.0 -0.5
[15] 10.0 -4.0 -0.5 10.0 -7.5  3.0  6.5
```

Sampling distribution under the sharp null



Probabilities under the sharp null

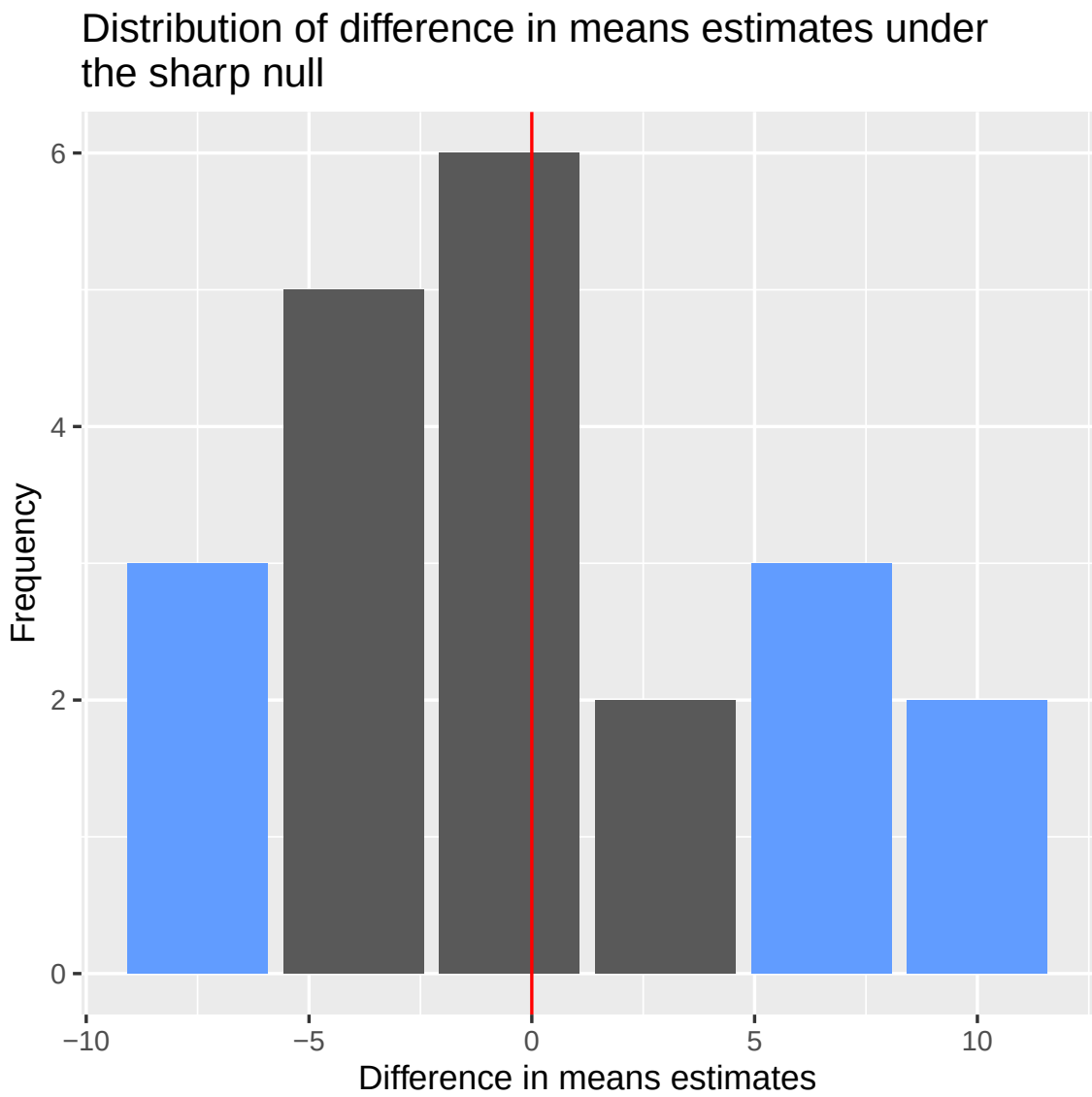
```
> prop.table(table(dm))
```

```
dm
     -7.5         -4       -0.5          3        6.5         10
0.1428571 0.2380952 0.2857143 0.0952381 0.1428571 0.0952381
```

```
> (pval ← mean(abs(dm) ≥ dm_hat))
```

```
[1] 0.3809524
```

Two-sided p-value



Calculating the p-value

```
> prop.table(table(dm))
```

```
dm
     -7.5         -4       -0.5          3        6.5         10
0.1428571 0.2380952 0.2857143 0.0952381 0.1428571 0.0952381
```

```
> (pval ← mean(abs(dm) ≥ dm_hat))
```

```
[1] 0.3809524
```

p-value

- Under the null distribution, if we were to re-randomize the experiment many times, we would see a value *at least as extreme as our estimate* 38.1% of the time.
- That doesn't seem **that** unlikely–more than one in three times, we would see an estimate as large as we got, just by chance.

- Why is this test called an "exact" test?
- Because we know the *exact* distribution of our estimate under the specified null. We do not have to approximate the distribution.

Note that we are conducting inference with respect to:

- a defined population
- a defined treatment
- a defined outcome
- a known treatment assignment mechanism
- a specified estimand
- a specified estimating procedure

- We could have used a different estimand
- Or a different estimator
- Randomization inference is also helpful because we can account for the nature of the treatment assignment procedure
  - For example, if treatment is assigned by clusters
  - Or matched pairs

The null

- How do we determine what the null is?
- We formalize our hypotheses in terms of the effect we are trying to find in the data. Is there a treatment effect? Is there a difference between these two groups?
- The null is (often, but not always) the case when there is no effect, or no difference.
- We can imagine other kinds of hypotheses, for example that effects are bounded away from zero and positive, or exactly .2. And we can characterize the distribution of our test statistic under the null.

The null

- Is the sharp null of no individual level effects plausible in this setting?
- Does it matter?

Different randomization procedures

(Athey and Imbens, 2017)

- Simple random assignment (Not discussed)
- Completely randomized experiments
- Stratified randomized experiments
- Paired randomized experiments
- Clustered randomized experiments (Why?)
  - SUTVA
  - What is the estimand?

Covariates

- Why we do balance tests
- More on this in Lecture 4

References I

Athey, S. and Imbens, G. W. (2017). The econometrics of randomized experiments. In Handbook of economic field experiments, volume 1, pages 73–140. Elsevier.

Butler, D. M. and Broockman, D. E. (2011). Do politicians racially discriminate against constituents? a field experiment on state legislators. American Journal of Political Science, 55(3):463–477.

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Holland, P. W. (1986). Statistics and causal inference. Journal of the American statistical Association, 81(396):945–960.